

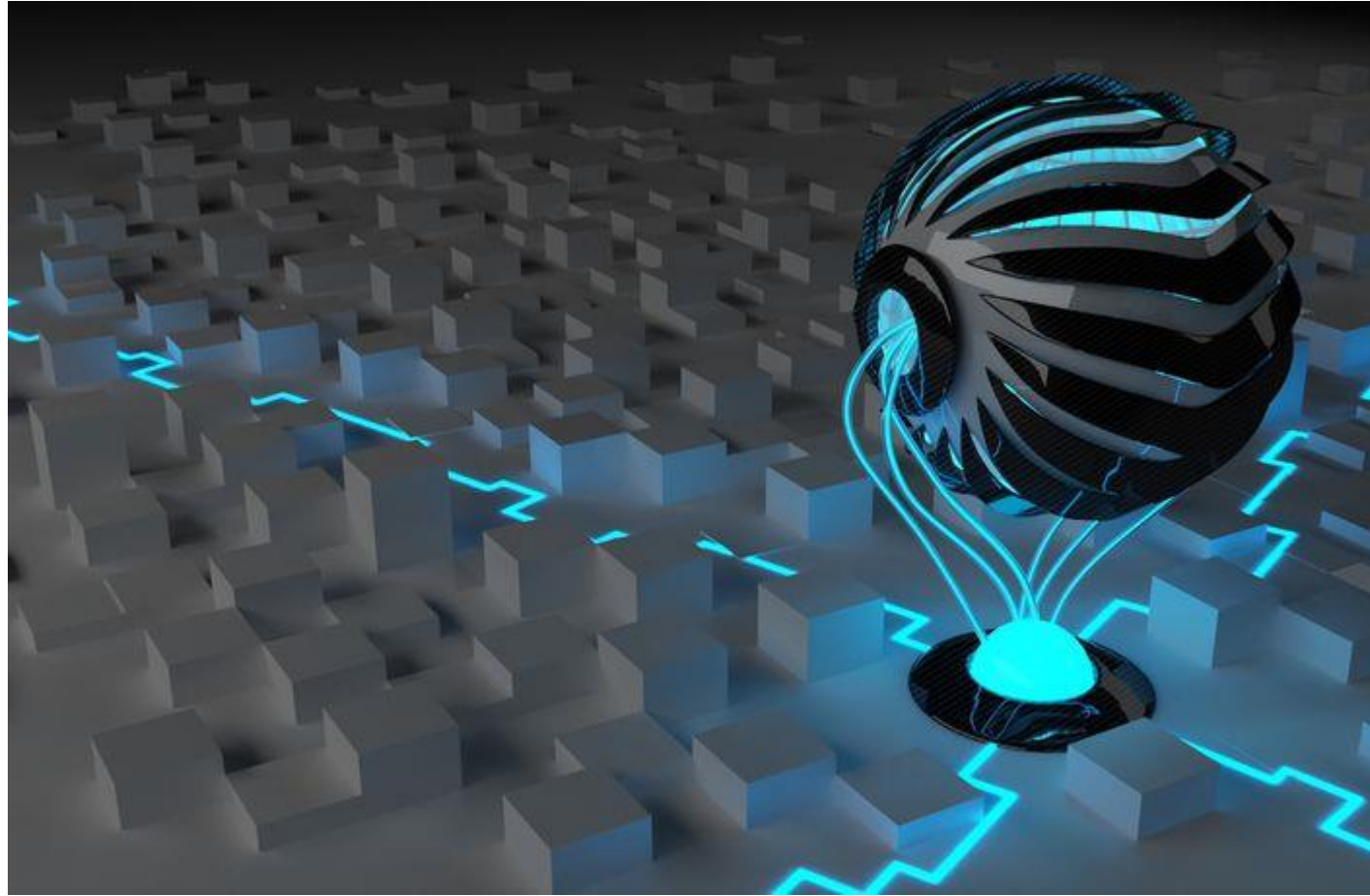
Computer Graphics and Visualization

Chapter 6

Illumination and Surface Rendering methods

Prepared by:

Asst. Prof. Sanjivan Satyal



Illumination and Surface Rendering methods [4 hours]

6.1. Algorithms to simulate ambient, diffuse, and specular reflection

6.2. Constant, Gouraud, Phong, and Fast Phong shading models

Illumination Models and Surface Rendering

- ❑ Once visible surface has been identified by hidden surface algorithm, a shading model is used to compute the intensities and color to display for the surface
- ❑ For realistic displaying of 3d scene it is necessary to calculate appropriate color or intensity for that scene
- ❑ In order to achieve realism in computer generated images, we need to apply natural lighting effects to the surfaces of objects
- ❑ **Illumination models** are used to calculate the amount of light reflected from a **certain position** on a surface

- ❑ An *illumination model* in computer graphics
 - ❑ Also called a *lighting model* or a *shading model*
 - ❑ Used to calculate the color of an illuminated position on the surface of an object
- ❑ Using the illumination model for computing the color of every pixel is costly. Therefore **surface rendering methods** are used to interpolate illumination values computed for certain special pixels (like vertices of a triangle).
- ❑ A *surface-rendering method* determine the pixel colors for all projected positions in a scene
- ❑ Photorealism in computer graphics involves
 - ❑ Accurate representations of surface properties, and
 - ❑ Good physical descriptions of the lighting effects

- Modeling the lighting effects that we see on an object is a complex process, involving principles of both physics and psychology
- **Physics:**
 - Material properties, light sources, relative positions, properties medium
- **Psychology:**
 - Perception, what do we see
 - Color
- Often approximating models

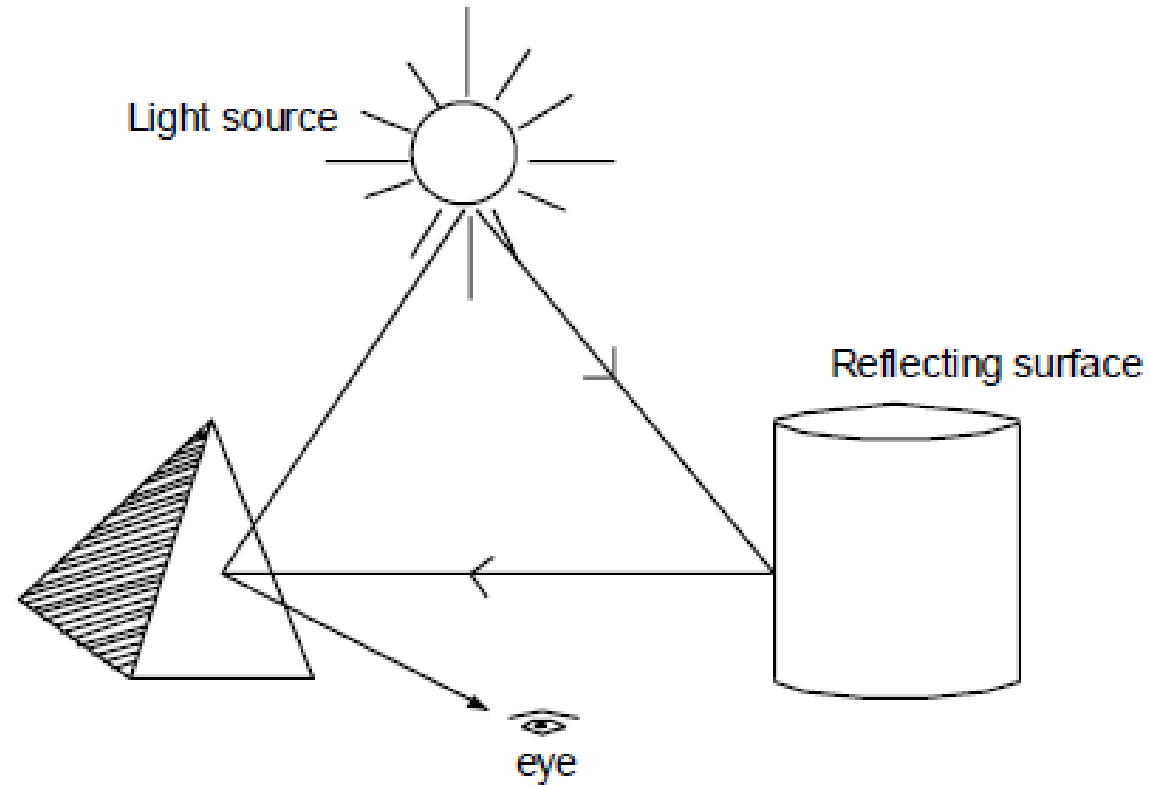


Figure : Using correct lighting improves the three-dimensional impression

Light Sources

- ❑ Object that radiates energy are called light sources, such as sun, lamp, bulb, fluorescent tube etc.
- ❑ Sometimes light sources are referred as light emitting object and light reflectors
- ❑ Generally light source is used to mean an object that is emitting radiant energy e.g. Sun
- ❑ **Total Reflected Light** = Contribution from light sources + contribution from reflecting surfaces

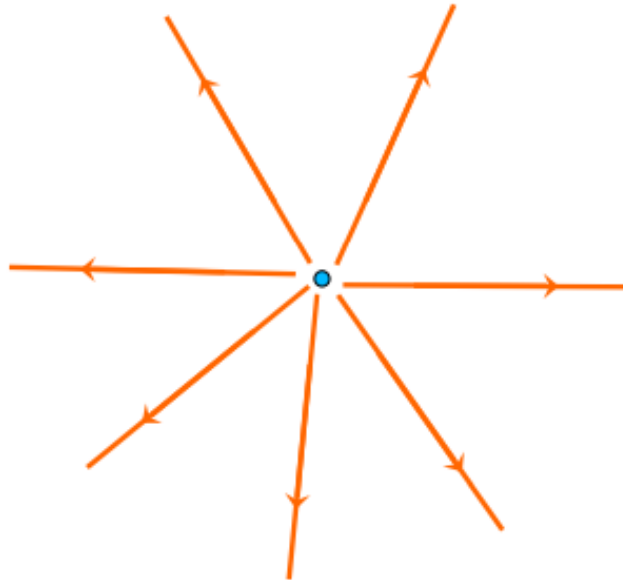
- ❑ A light source can be defined with the following parameters:
 - ❑ position, color of the emitted light, emission direction, shape of the light source
- ❑ Often in computer graphics, light sources are assumed as simple points
 - ❑ it is easier to do all the necessary lighting computations with point light sources



Light Sources

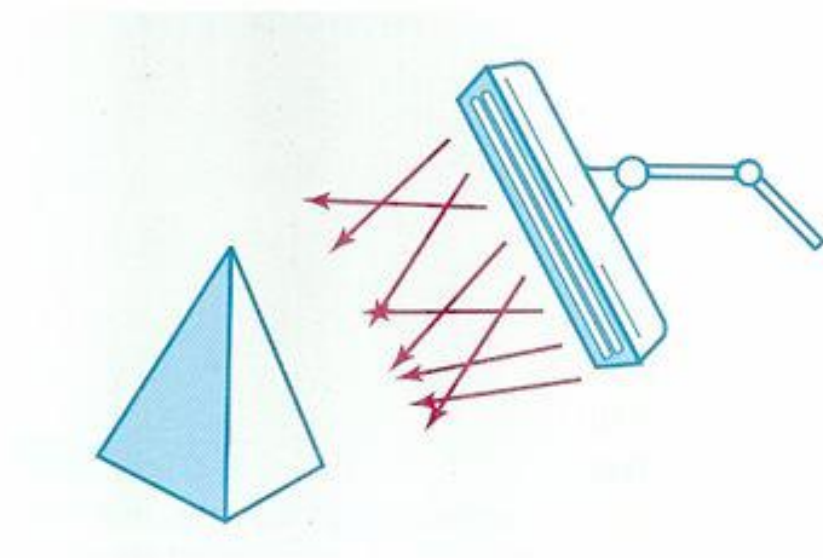
1.Point Source:

- Point source is the simplest light emitter e.g. light bulb. When light source model is a reasonable approximation for sources whose dimensions are small compared to the size of objects in a scene



2. Distributed Light Source:

- Area of source is not small compared to the surfaces in the scene e.g. Fluorescent lamp



Distributed Light Source

Radial Intensity Attenuation

- ❑ As radiant energy from a point light source travels through space, its amplitude is attenuated by the factor $1/d^2$, where d is the distance that the light has traveled.
- ❑ This means that a surface close to the light source (small d) receives higher incident intensity from the source than a distant surface (large d).
- ❑ Therefore to produce realistic lighting effects, illumination model should take intensity attenuation into account. Otherwise we are likely to illuminate all surfaces with same intensity.

- For a point light source attenuation factor is $1/d^2$.
- And for a distributed light source attenuation factor is given by **inverse quadratic attenuation function**

$$f_{RadAtten}(d_l) = \frac{1}{a_0 + a_1 d_l + a_2 d_l^2}$$

- For light sources, that are located very far away from the objects of the scene it is safe to assume that there is no attenuation, i.e. $f_{RadAtten} = 1.0$.

Intensity Attenuation

- We cannot apply intensity-attenuation calculation 1 to a point source at "infinity," because the distance to the light source is indeterminate. Also, all points in the scene are at a nearly equal distance from a far-off source.
- To accommodate both remote and local light sources, we can express the intensity-attenuation function as

$$f_{l,\text{radatten}} = \begin{cases} 1.0, & \text{if source is at infinity} \\ \frac{1}{a_0 + a_1 d_l + a_2 d_l^2}, & \text{if source is local} \end{cases}$$

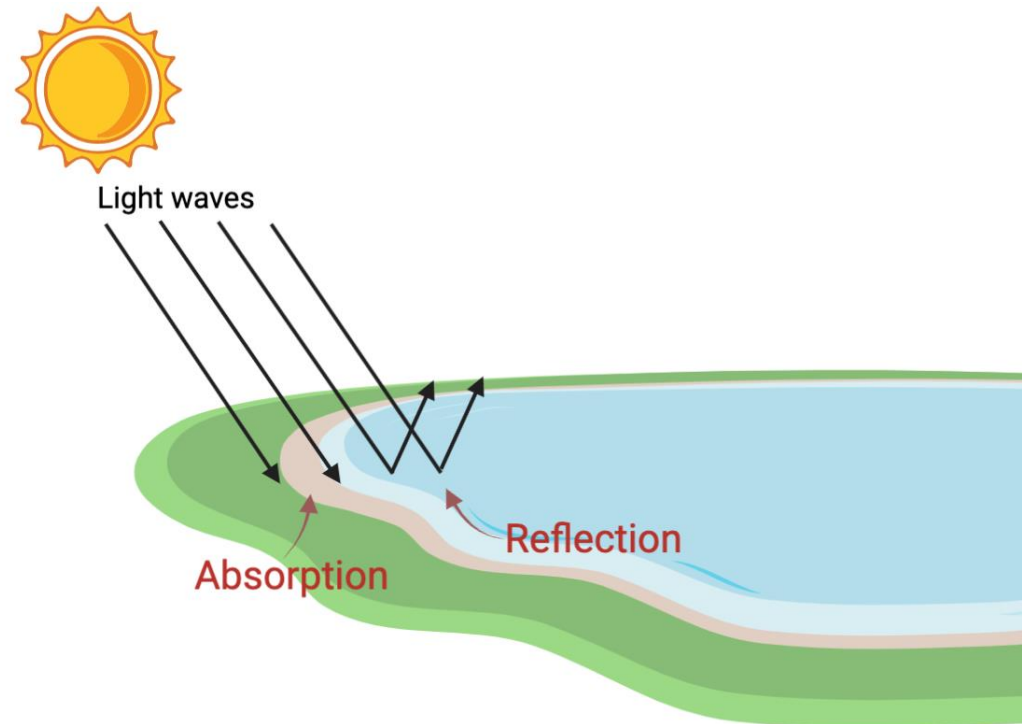
Surface Lighting Effects

- ❑ The light reflected from the surface of an object depends on several factors
 - ❑ Degree of reflectivity of the object (shiny vs. dull surfaces)
 - ❑ Shining material reflects more incident light
 - ❑ Dull surface absorbs more of the incident light
 - ❑ Surface texture properties
 - ❑ Color reflectance coefficients (e.g. color of the object)
- ❑ For transparent surfaces, some of the incident light will be reflected and some will be transmitted through the material

Reflection of light

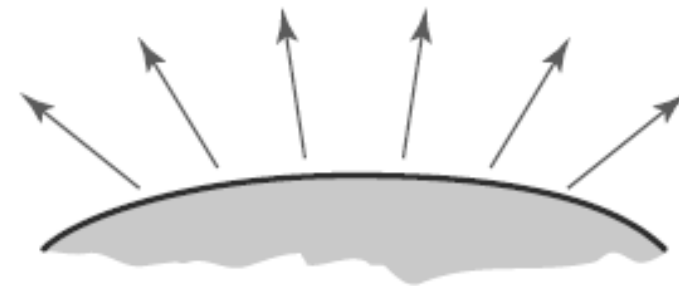
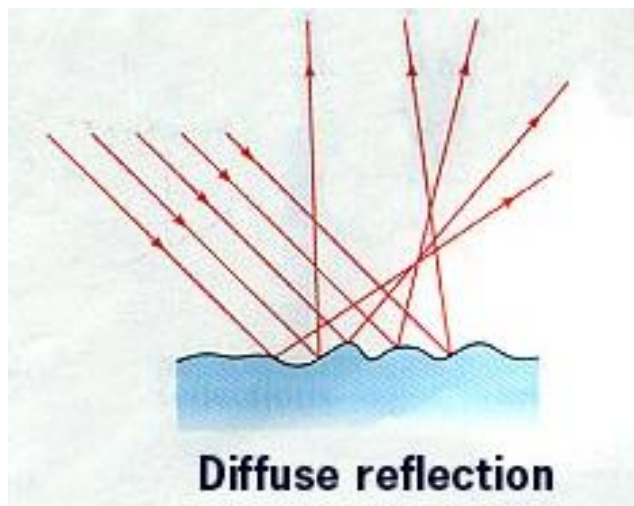
When light is incident on opaque surface part of it is reflected and part of it is absorbed.

$$\therefore I = A + R$$



Diffuse reflection

- ❑ Surface that are rough or grainy, tend to scatter the reflected light in all direction which is called diffuse reflection
- ❑ Surfaces appear equally bright from all viewing angles since they reflect light with equal intensity in all directions. E.g. rough, matte surfaces
- ❑ Position of viewer is not important

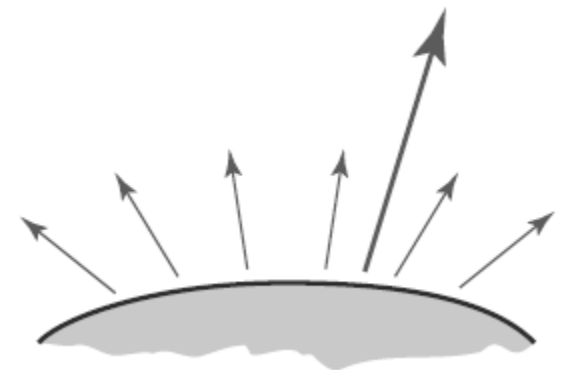
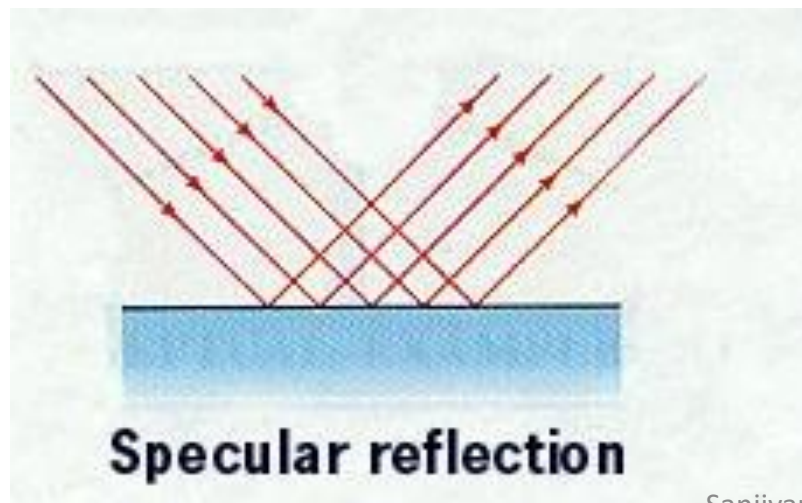


Diffuse reflection...

- ❑ Color of an object is determined by the color of the diffuse reflection of the incident light
- ❑ When an object is illuminated with white light the original color of the object is what we see as the diffuse reflection
 - ❑ If a blue object is illuminated with red light, it will appear black
 - ❑ If any object surface is red then there is a diffuse reflection for red component of light and all other components are absorbed by the surface

Specular reflection

- In addition to diffuse light scattering, some of the reflected light is concentrated into a highlight, or bright spot, called **specular reflection**.
- Light reflected with unequal intensity. E.g. shiny surface. polished metal, an apple, or a person's forehead, only when we view the surface from a particular direction.



Diffuse vs. Specular Reflection



Diffuse Reflection



Diffuse and Specular
Reflection

Illumination models

- ❑ Illumination models are used to calculate light intensities that we should see at a given point on the surface of an object
- ❑ Lighting calculations are based on the optical properties of surfaces, the background lighting conditions and the light source specifications
- ❑ All light sources are considered to be point sources, specified with a co-ordinate position and an intensity value (color). Some illumination models are:

1. Ambient Light
2. Diffuse Reflection
3. Specular Reflection and Phong model

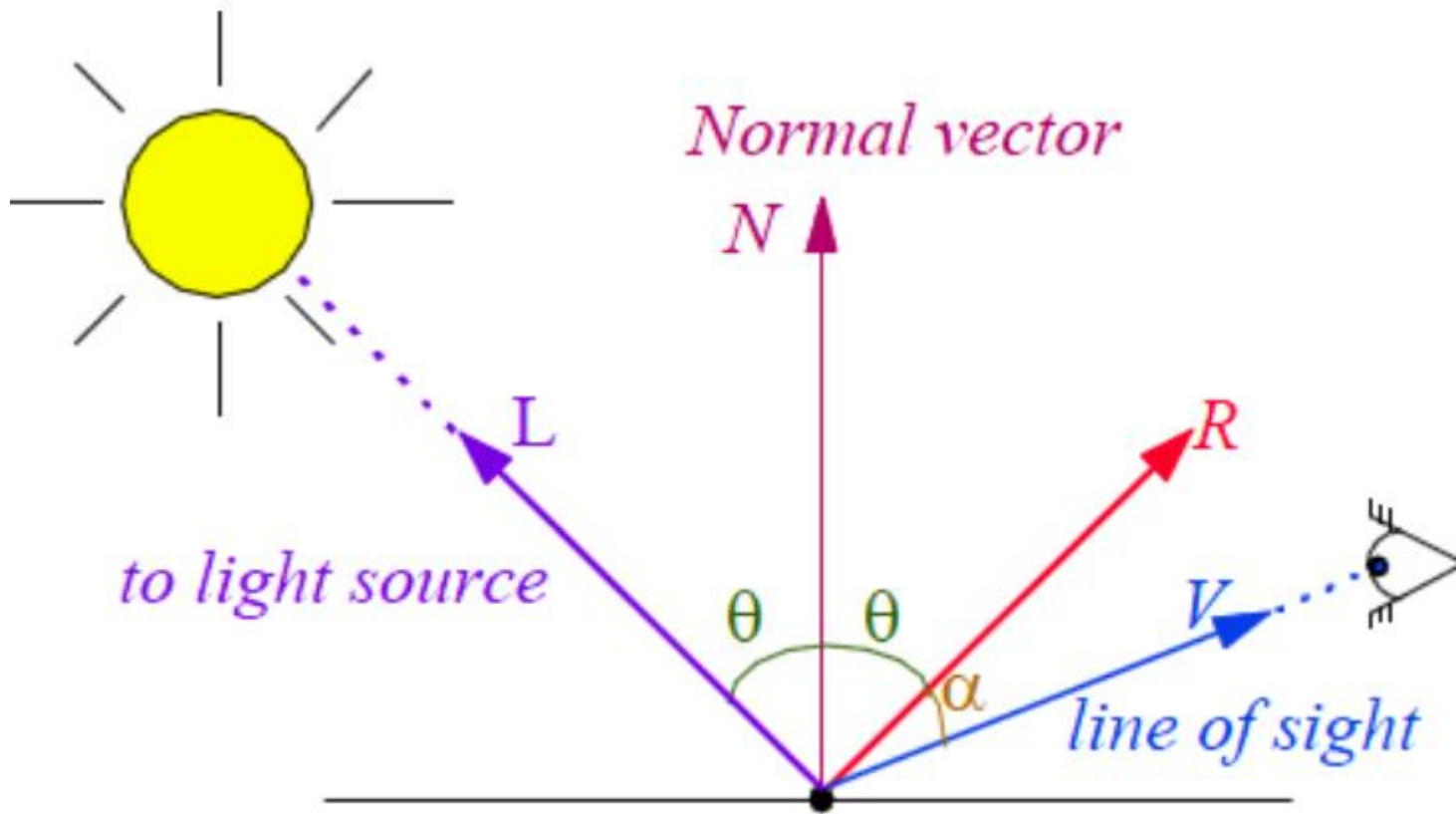
Illumination models...

- Ignoring color for now: the necessary computations are the same for each source and for each primary color

$$I = \underbrace{k_a L_a}_{I_a} + \underbrace{k_d L_d}_{I_d} + \underbrace{k_s L_s}_{I_s}$$

- This means the surface shade is a function of L_a ambient light L_d diffuse light L_s specular light

Useful Concept



L - light source direction
 N - Normal vector
 θ - angle of incidence
 V - line of sight
 R - direction of ideal specular reflection
 α - angle between R and V

1.Ambient light

- ❑ Surface getting light from various reflected source i.e. light not coming directly from a light source but coming after getting reflected from other surface. Also called **background light**
- ❑ Constant color reflected from all points on the surface, ambient light produce constant illumination
- ❑ Tells us how bright the surface will look like when no light source can directly reach the surface
- ❑ When used alone, does not produce very interesting pictures



- Ambient light has no spatial or directional characteristics and amount on each object is a constant for all surfaces and all directions
- Every object in the scene will be illuminated by the amount independent of the surface orientation and viewer location
- But different surfaces may reflect different amount of ambient light based on their absorbance/reflectance properties. We can model this by a constant factor for each surface:

$$I = I_a K_a$$

- Where I is the resulting intensity and I_a is the amount of ambient light incident on any surface and K_a is ambient reflection coefficient
- If we assume that ambient light impinges equally on all surface from all direction, then

$$I = I_a K_a$$

- Where I_a is intensity of incident ambient light. The amount of light reflected from an object's surface is determined by K_a , the ambient-reflection coefficient. K_a ranges from 0 to 1.

2. Diffuse Reflection

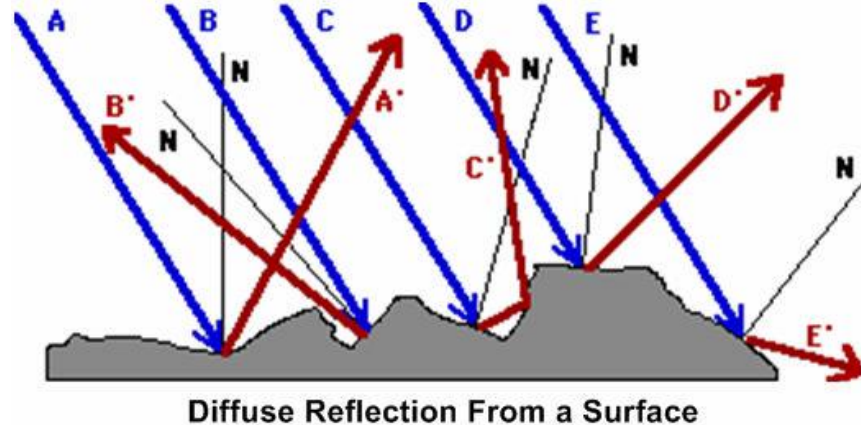
- ❑ Objects illuminated by ambient light are uniformly illuminated across their surfaces even though light are more or less bright in direct proportion of ambient intensity
- ❑ The light that is reflected in **all directions** is called diffuse reflection
- ❑ The reflected light is independent of the viewing position (equally bright from all viewing directions)
- ❑ But the light position with respect to the **surface orientation** is important to determine the light reflected from the surface

Diffuse Reflection...

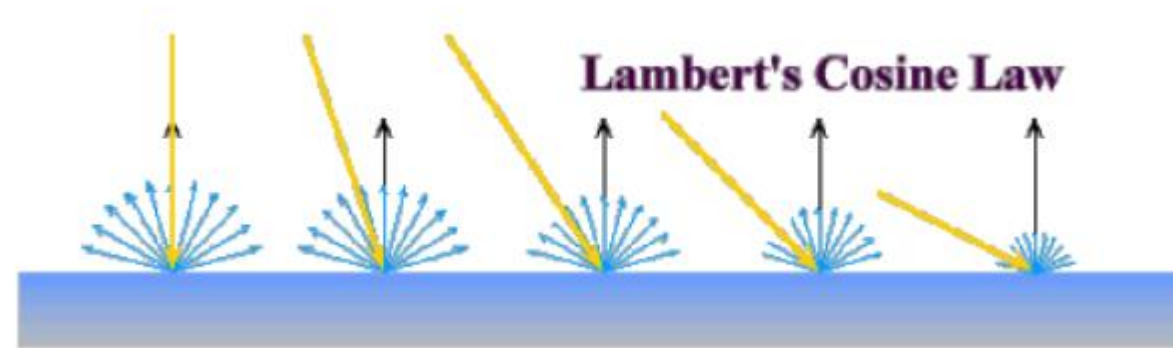
- ❑ The **diffuse-reflection coefficient**, or **diffuse reflectivity**, k_d (varying from 0 to 1) define the fractional amount of the incident light that is diffusely reflected
- ❑ The surface parameter k_d (actually function of surface color) depends on the reflecting properties of material **so for highly reflective surfaces, the k_d nearly equal to 1.**
- ❑ If a surface is exposed only to ambient light, we can express the intensity of the diffuse reflection at any point on the surface as:
- ❑ **$I_{amb,Diff} = k_d \cdot I_a$** , (where I_a = intensity of ambient light)

ideal diffuse reflectors

- ❑ An ideal diffuse surface is, at the microscopic level, a very rough surface.
Example: chalk, clay, some paints
- ❑ Incident light is scattered with equal intensity in all directions
- ❑ Surfaces appear equally bright from all direction
- ❑ also referred to as **Lambertian reflectors** as they are governed by the **Lambert's cosine law**

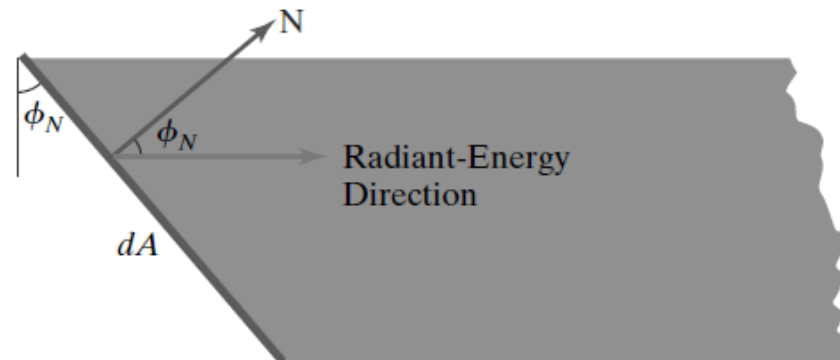


- Ideal diffuse reflectors reflect light according to Lambert's cosine law, Lambert's law states that the reflected energy from a small surface area in a particular direction is proportional to cosine of the angle between that direction and the surface normal



This law states that the amount of radiant energy coming from any small surface area dA in a direction ϕ_N relative to the surface normal is proportional to $\cos \phi_N$

$$E \propto \cos \theta \quad \dots (1)$$



Explanation

- **Case I** – Consider a surface that is normal to the luminous flux as shown in Figure-1.

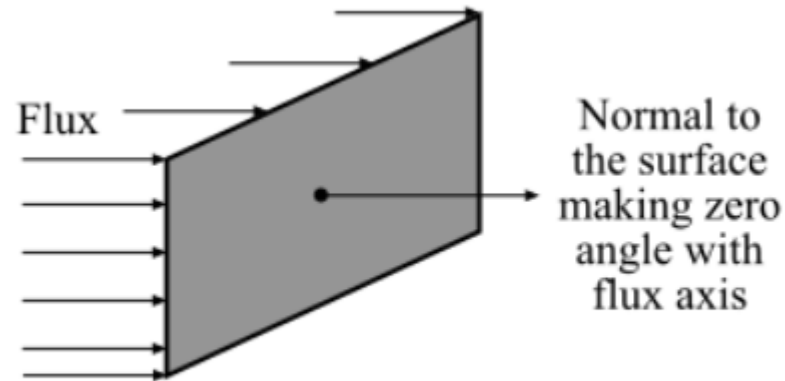


Figure 1 - Normal Surface

Then, as per Lambert's cosine law, we get,

$$\text{Illumination, } E = \frac{\phi}{A} \dots (2)$$

- **Case II** – Consider a surface inclined to the flux in such a way that the normal to the surface makes an angle (θ) to the flux axis, as shown in Figure-2.

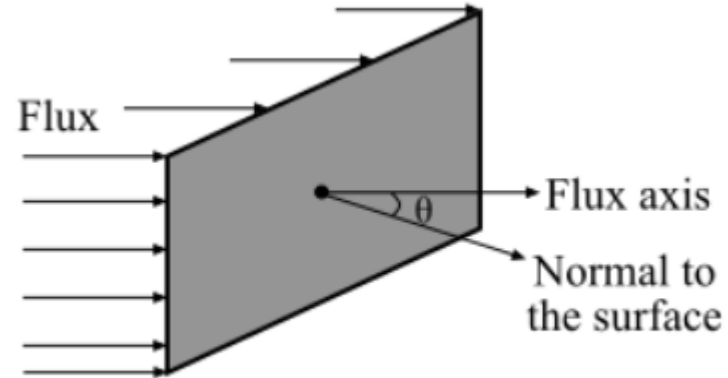


Figure 2 - Inclined Surface

Then, by the Lambert's cosine law, we get,

$$E = \frac{\phi}{A} \times \cos \theta \quad \dots (3)$$

- **Case III** - Consider a point 'P' on a plane surface and the distance between the source of light (S) and the point 'P' is 'r' meters. The source (S) is located at height of 'h' meters from the surface and its luminous intensity is 'I' candle power as shown in Figure-3.

Then, by the cosine law of illumination, we get, the illumination at point 'P' as,

$$E_P = \frac{I}{r^2} \times \cos \theta$$

From Figure-3, we have,

$$\cos \theta = \frac{h}{r}$$

$$\Rightarrow r = \frac{h}{\cos \theta}$$

Therefore, the illumination at point 'P' is,

$$E_P = \frac{I}{(h/\cos \theta)^2} \times \cos \theta$$

$$\therefore E_P = \frac{I}{h^2} \times \cos^3 \theta \quad \dots (4)$$

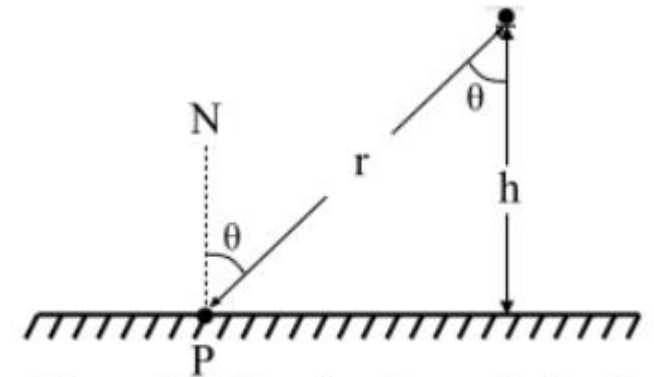


Figure 3 - Illumination at Point P

Where, I/h^2 is the illumination at any point located directly below the source of light.

Ambient + Diffuse Reflection

- If I_l , is the intensity of the point light source, then the diffuse reflection equation for a point on the surface can be written as

$$I_{l,diff} = k_d \cdot I_l \cos \theta$$

- A surface is illuminated by a point source only if the angle of incidence is in the range 0° to 90° ($\cos \theta$ is in the interval from 0 to 1)
- At any surface position, we can denote the unit normal vector as N and the unit direction vector to a point source as L , Then, $\cos \theta = N \cdot L$
- and the diffuse reflection equation for single point-source illumination is

$$I_{l,diff} = k_d \cdot I_l \cos \theta = k_d I_l (N \cdot L)$$

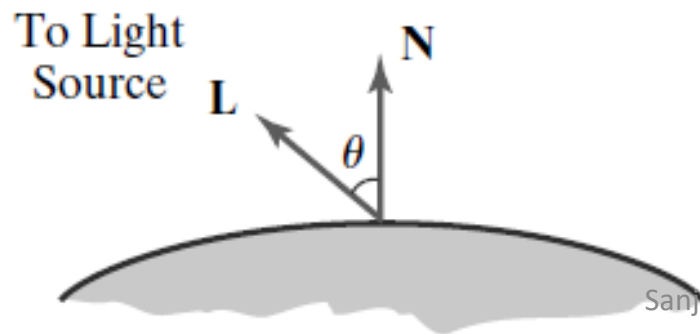


Figure: Angle of incidence between the unit light-source direction vector L and the Unit surface normal N

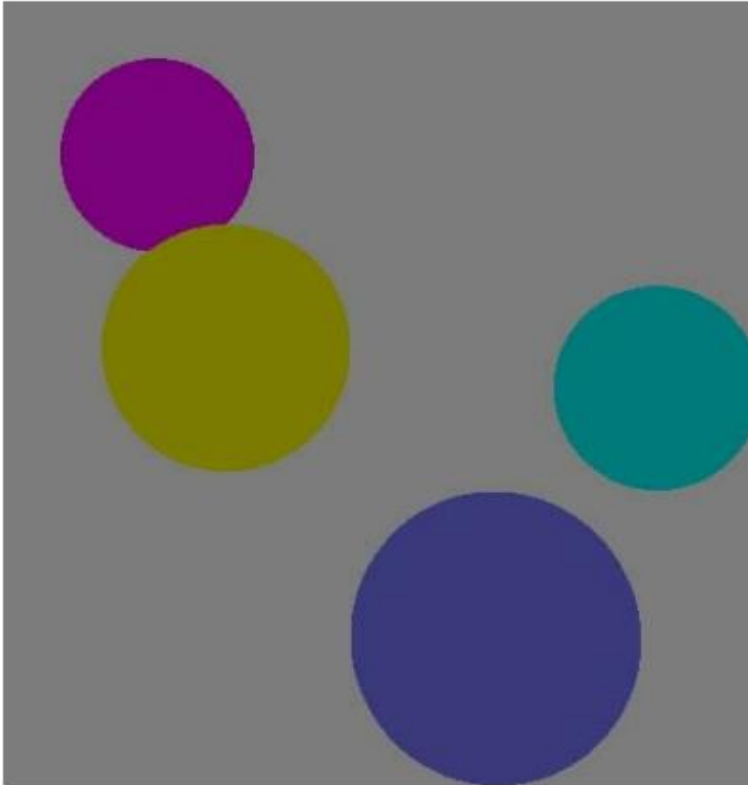
- We can combine the ambient and point source intensity calculations to obtain an expression for the **total diffuse reflection**. Thus, we can write the total diffuse reflection equation as:

$$I_{\text{diff}} = k_a \cdot I_a + k_d I_l (N \cdot L)$$

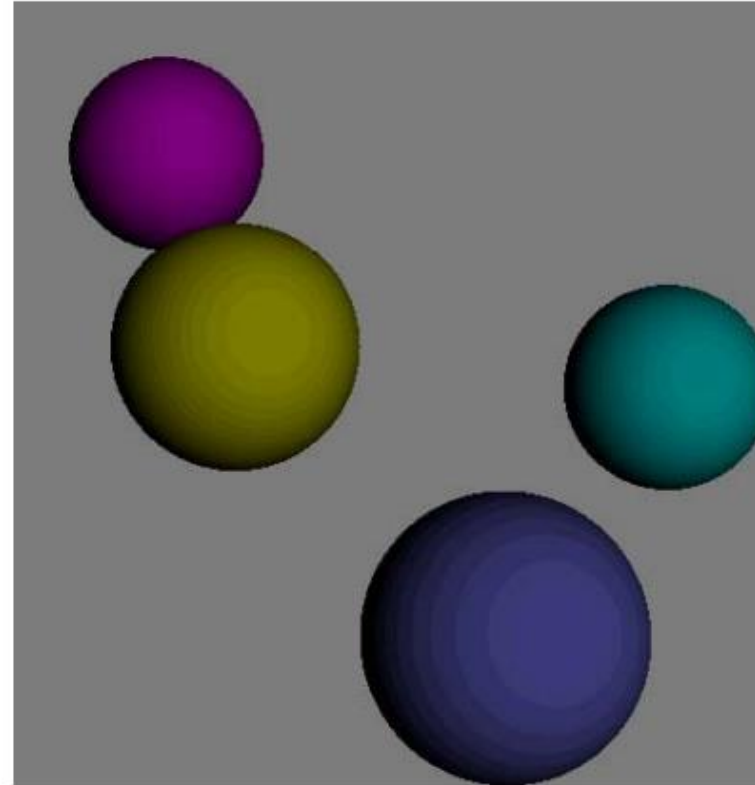
where both k_a (ambient-reflection coefficient) and k_d depend on surface material properties and are assigned values in the **range from 0 to 1**

$$I = \begin{cases} k_a I_a + k_d I_l (N \cdot L) & \text{if } N \cdot L > 0 \\ k_a I_a & \text{if } N \cdot L \leq 0 \end{cases}$$

Ambient Light vs. Diffuse Reflection



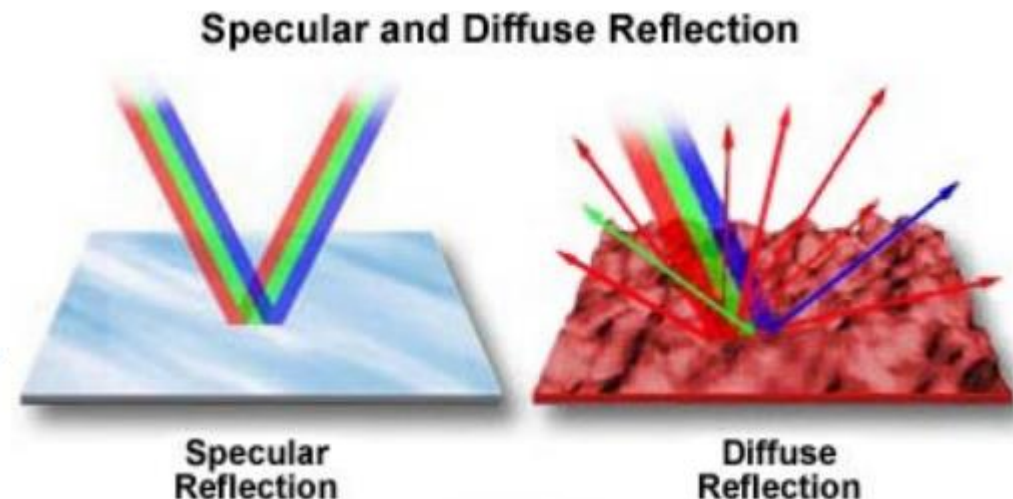
Ambient lighting only

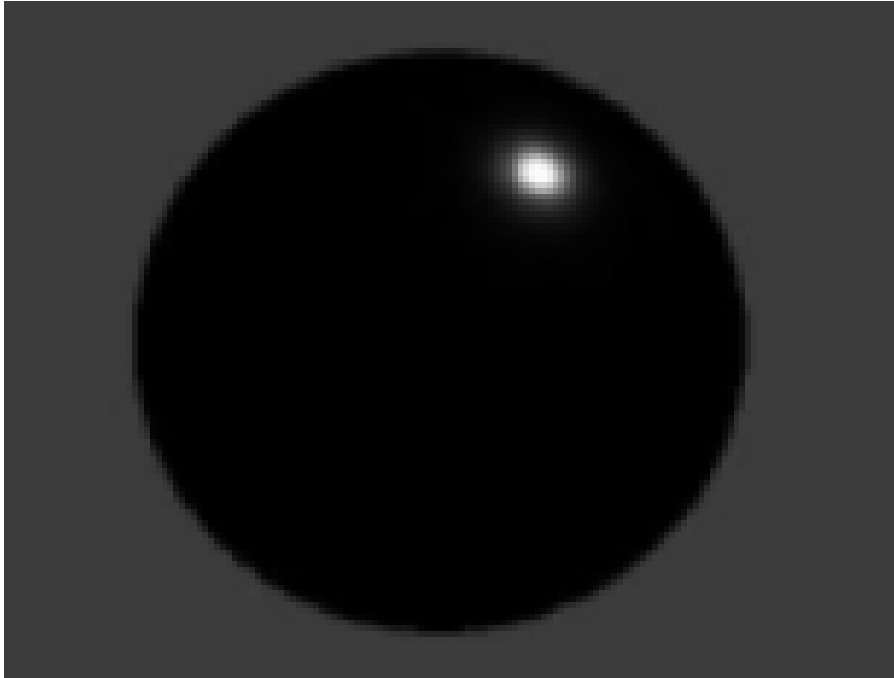


With diffuse lighting

3. Specular reflection

- When we look at an illuminated shiny surface, such as polished metal, a person's forehead, we see a highlight or bright spot, at certain viewing direction
- Light reflected from a certain spot on the object is concentrated and appears as a lot brighter compared to other spots. This is due to **specular reflection** and is an important property of shiny objects.
- Specular reflection is both dependent on the **light direction**, **surface orientation**, and **viewer position**





Specular reflection



Specular reflection

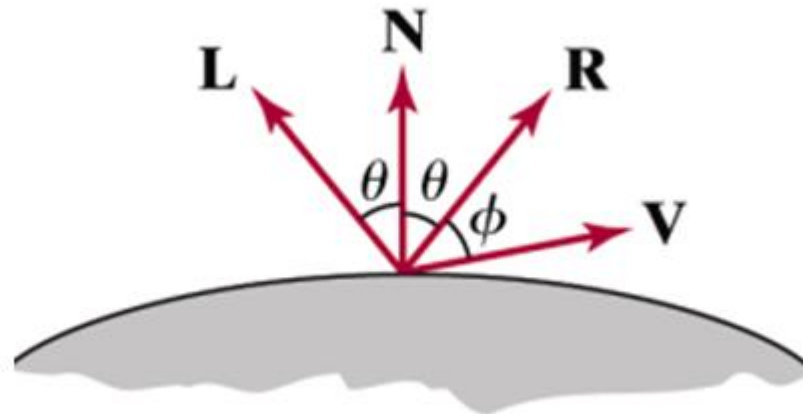


Diffuse reflection

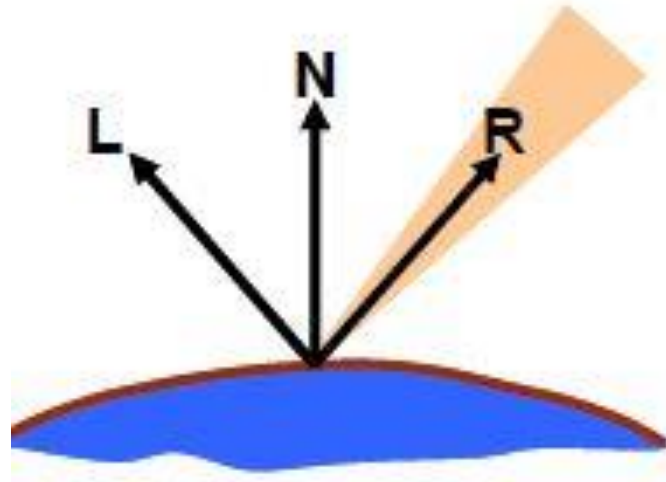
Specular reflection

Let SR (specular reflection angle) = angle of incidence as in figure.

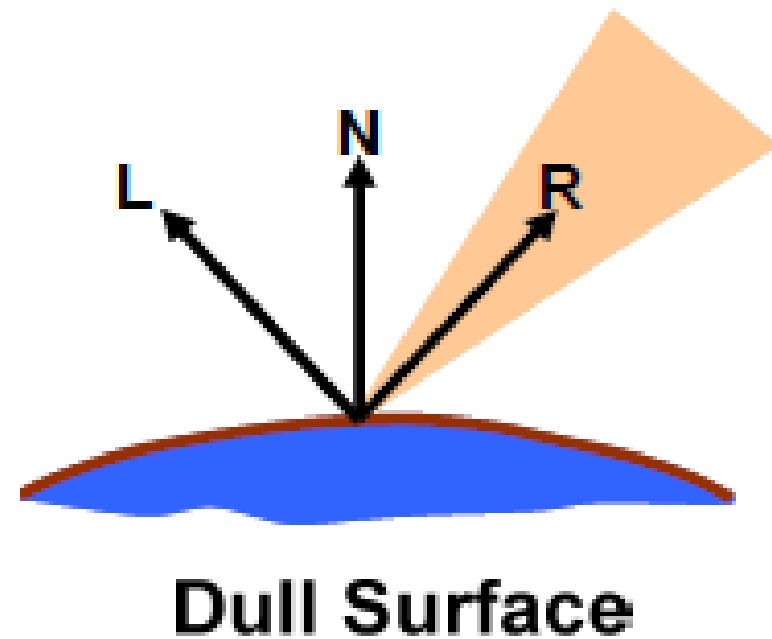
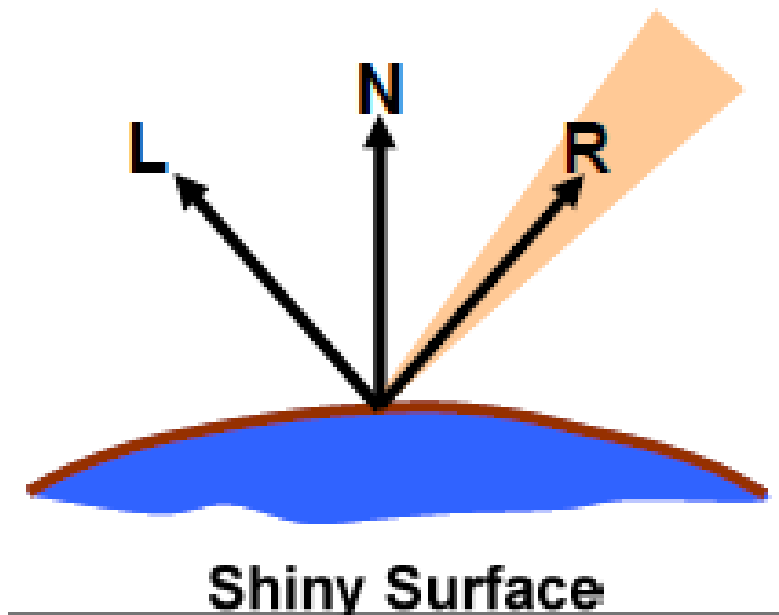
- N - unit vector normal to surface at incidence point
- R - unit vector in the direction of ideal specular reflection
- L - unit vector directed towards point light source
- V - unit vector pointing to the viewer from surface
- ϕ - the viewing angle relative to the specular reflection direction



- The bright spot, or specular reflection, that we can see on a shiny surface is the result of total, or near total, reflection of the incident light in a concentrated region around the **specular-reflection angle**.
 - "specular reflection angle = angle of incidence"
- The specular reflection is visible only at directions close to R
- Shiny surfaces like mirrors have a narrow specular reflection range
- **ideal reflector** (mirror) reflects all lights to the direction where angle of reflection is identical to the angle of incidence
 - V and R coincide ($\theta = 0$)



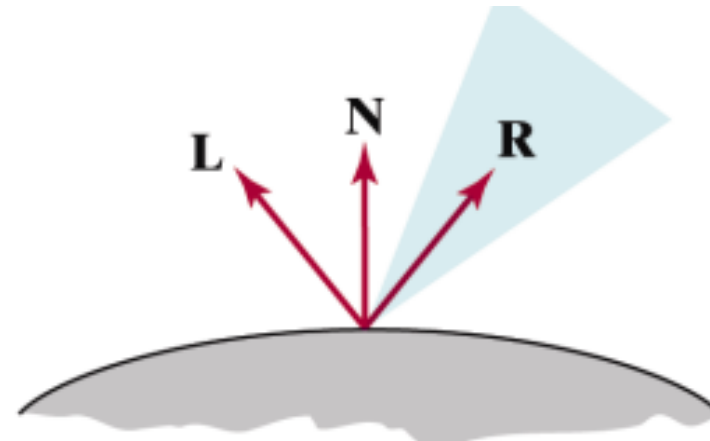
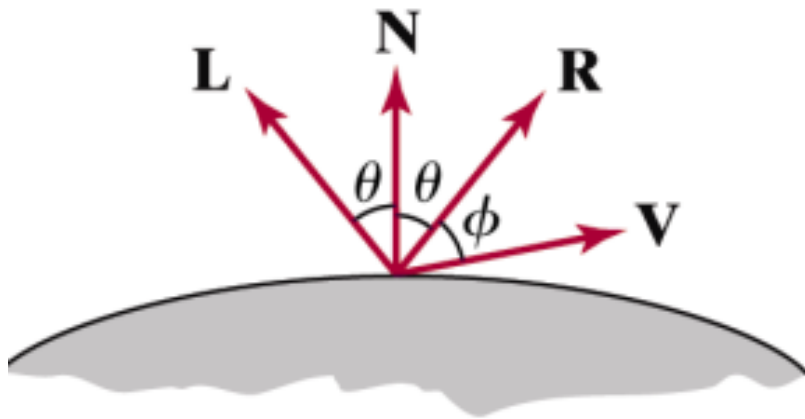
- Objects other than ideal reflectors exhibit specular reflections over a finite range of viewing positions around vector $\mathbf{R}$
- Shiny surface have a narrow specular-reflection range, and dull surfaces have a wider reflection range.



Phong Specular reflection (Assignment)

How much light is reflected?

- Depends on the angle θ between the ideal reflection direction R and the viewer direction V
- An empirical model for calculating the specular reflection range, developed by Phong Bui Tuong and called the **Phong specular-reflection model** or simply the **Phong model**



❑ For an ideal reflector (perfect mirror), incident light is reflected only in the specular-reflection direction. In this case, we would only see reflected light when vectors V and R coincide ($\theta = 0$)

❑ Phong model, sets the intensity of specular reflection proportional to

$$I_s \cos^n \theta$$

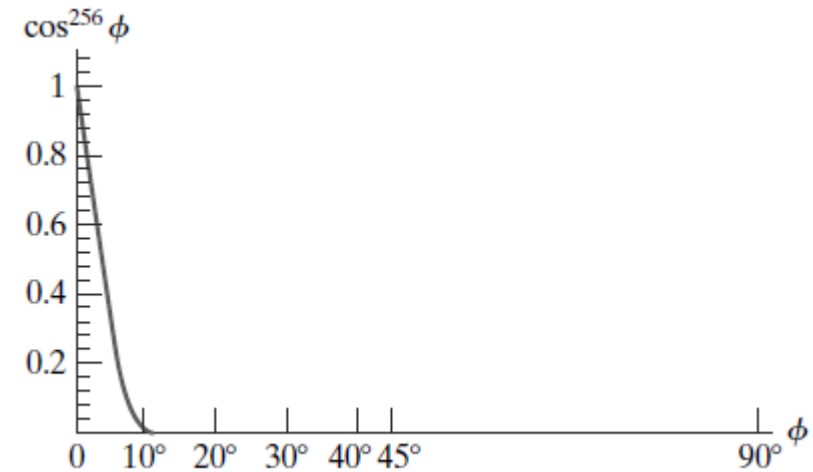
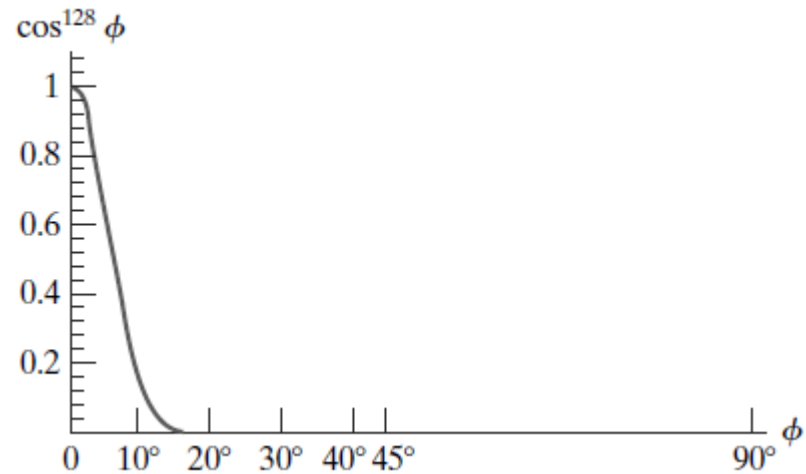
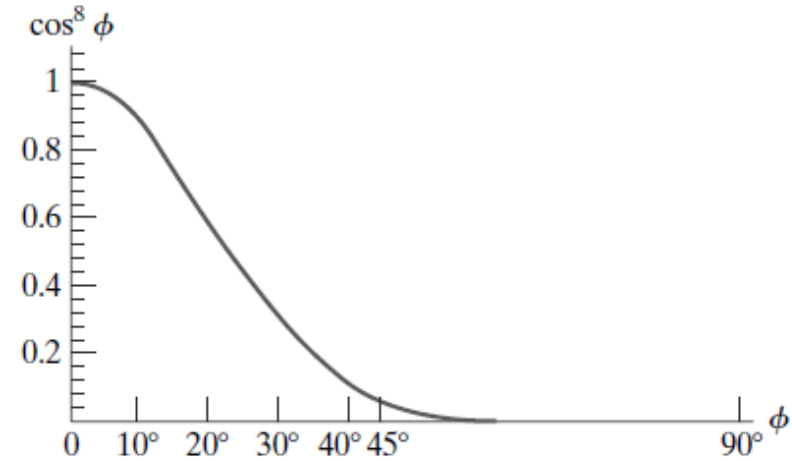
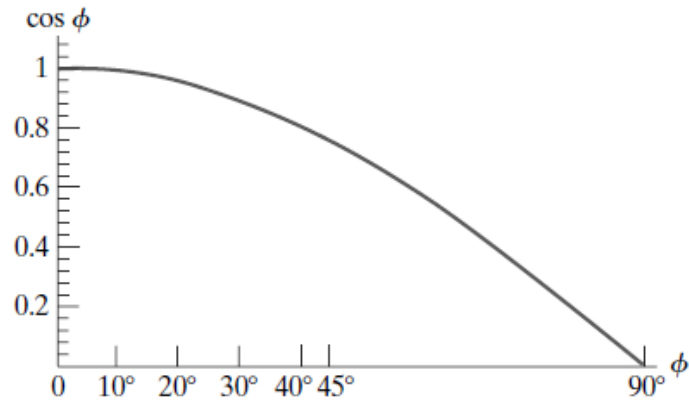
- n is specular reflection parameter (depends on surface)
 - θ ranges from 0 to 90 ($\cos \theta$ varies from 0 to 1)
-
- Intensity of specular reflection depends on:
 - Material properties of surface
 - Angle of incidence θ
 - Other factors such as polarization and color of the incident light

- ❑ A very shiny surface is modeled with a large value for n_s (say, 100 or more), and smaller values (down to 1) are used for duller surfaces
- ❑ For a perfect reflector, n_s is infinite
- ❑ Monochromatic specular intensity variations can be approximated using specular-reflection coefficient, $W(\theta)$ for each surface, we can write the Phong specular-reflection model as

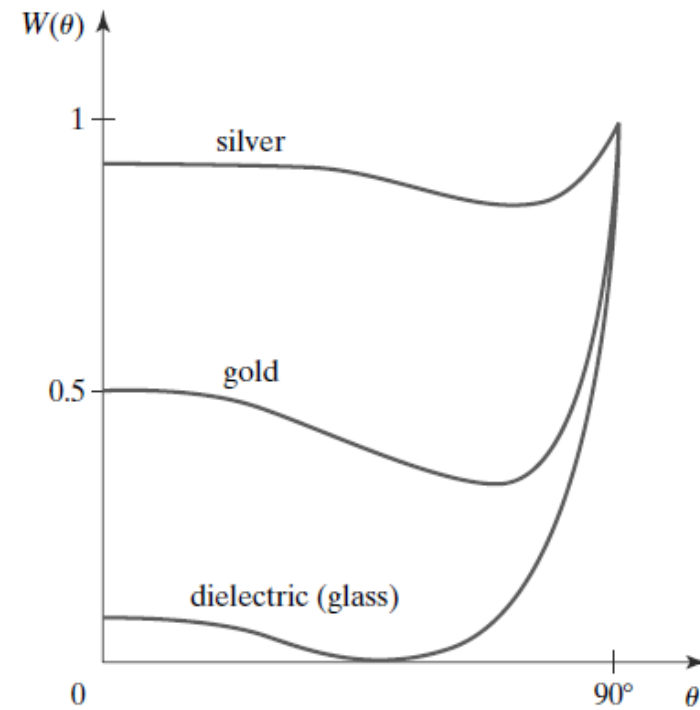
$$I_{spec} = w(\theta) I_l \cos^{n_s} \phi$$

- I_l = intensity of the light source,
- ϕ is the viewing angle relative to the specular-reflection direction $\mathbf{R}$.
- At $\theta = 90$, $w(\theta) = 1$, all incident light is reflected

Plots of $\cos n_s \phi$ using different values for the specular exponent n_s



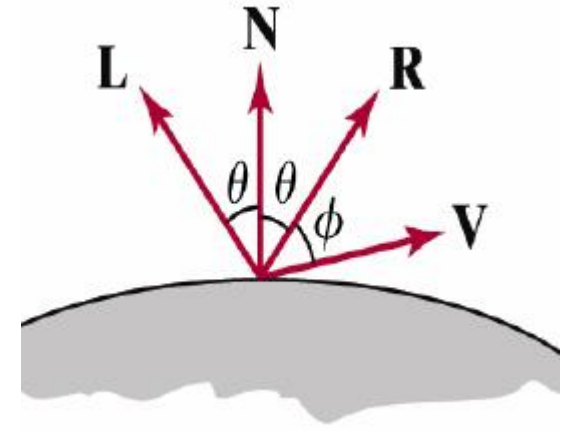
- Approximate variation of the specular-reflection coefficient for different materials, as a function of the angle of incidence.



- ❑ Because $\mathbf{V}$ and $\mathbf{R}$ are unit vectors in the viewing and specular-reflection directions, we can calculate the value of $\cos \theta$ with the dot product $\mathbf{V} \cdot \mathbf{R}$.
- ❑ In addition, no specular effects are generated for the display of a surface if $\mathbf{V}$ and $\mathbf{L}$ are on the same side of the normal vector $\mathbf{N}$ or if the light source is behind the surface.
- ❑ Thus, assuming the specular-reflection coefficient is a constant for any material, replace $w(\theta)$ with k_s .
- ❑ Then, we can determine the intensity of the specular reflection due to a point light source at a surface position with the calculation

$$I_{l,\text{spec}} = \begin{cases} k_s I_l (\mathbf{V} \cdot \mathbf{R})^{n_s}, & \text{if } \mathbf{V} \cdot \mathbf{R} > 0 \quad \text{and} \quad \mathbf{N} \cdot \mathbf{L} > 0 \\ 0.0, & \text{if } \mathbf{V} \cdot \mathbf{R} \leq 0 \quad \text{or} \quad \mathbf{N} \cdot \mathbf{L} \leq 0 \end{cases}$$

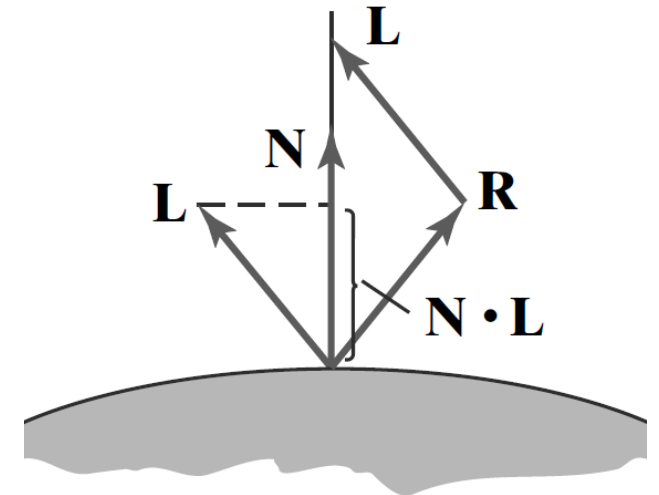
- ❑ The direction for $\mathbf{R}$, the reflection vector, can be computed from the directions for vectors $\mathbf{L}$ and $\mathbf{N}$.
- ❑ As seen in Figure, the projection of $\mathbf{L}$ onto the direction of the normal vector has a magnitude equal to the dot product $\mathbf{N} \cdot \mathbf{L}$, which is also equal to the magnitude of the projection of unit vector $\mathbf{R}$ onto the direction of $\mathbf{N}$.



- ❑ Therefore, from this diagram, we see that

$$\mathbf{R} + \mathbf{L} = (2\mathbf{N} \cdot \mathbf{L})\mathbf{N}$$

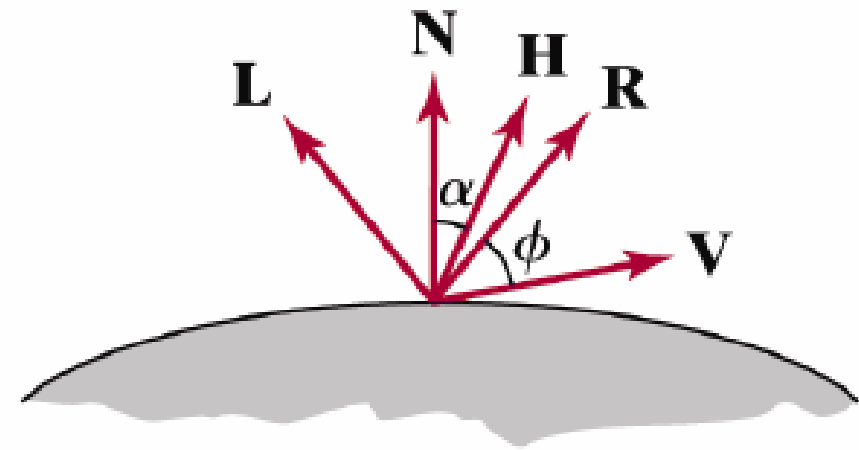
$$\mathbf{R} = (2\mathbf{N} \cdot \mathbf{L})\mathbf{N} - \mathbf{L}$$



Simplified Phong model using halfway vector

- Replacing $V \cdot R$ with $N \cdot H$ where H is halfway vector between L and V (i.e. H is unit bisector vector of angle between L and V)
 - this simply replaces the empirical $\cos \theta$ calculation with the empirical $\cos \alpha$ calculation
 - H is constant if both viewer and the light source are sufficiently far from the surface
- The halfway vector is obtained as

$$H = \frac{L + V}{|L + V|}$$



Combined Diffuse and Specular Reflection

- Point Light Source

$$\begin{aligned} I &= I_{diff} + I_{spec} \\ &= k_a I_a + k_d I_l (N \cdot L) + k_s I_l (N \cdot H)^{n_s} \end{aligned}$$

- Multiple light sources (n light sources)

$$I = k_a I_a + \sum_{i=1}^n I_{li} \left[k_d (N \cdot L_i) + k_s (N \cdot H_i)^{n_s} \right]$$

- The Phong illumination model considering attenuation is:

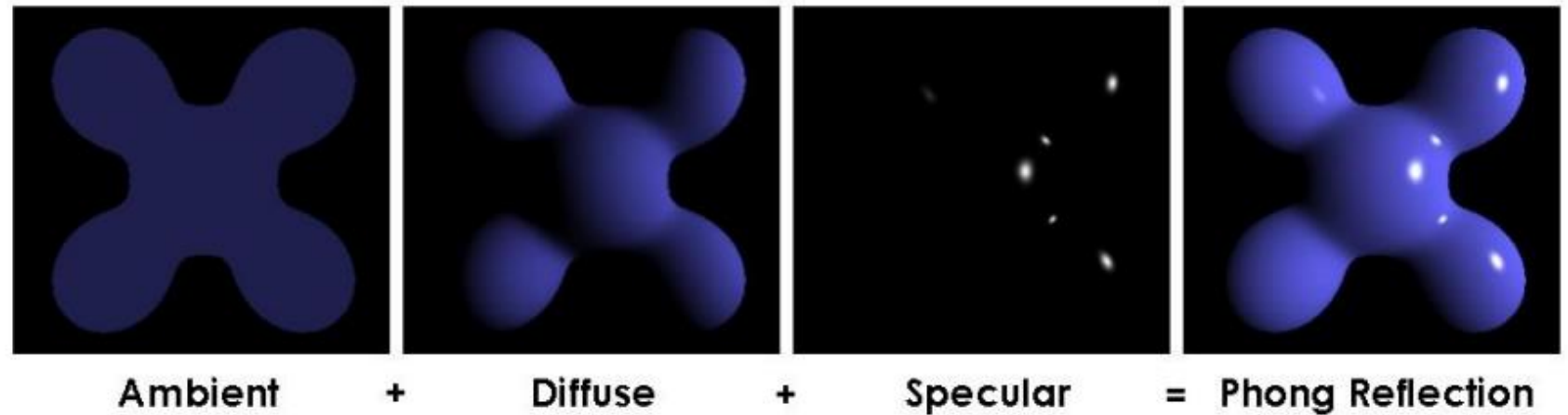
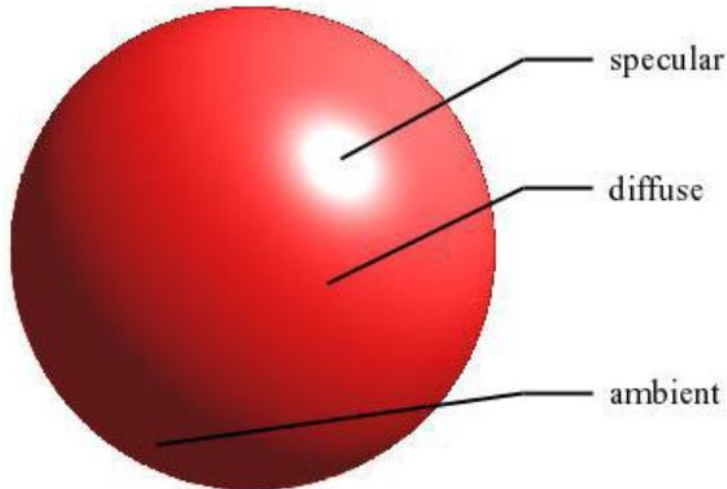
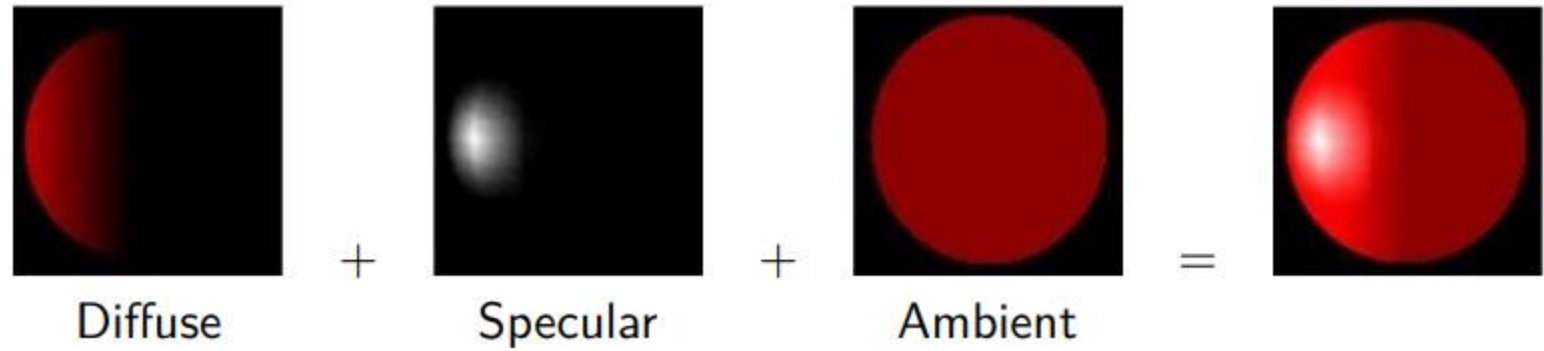
$$I = k_a I_a + \sum_{i=1}^n f(d_i) I_{li} \left[k_d (N \cdot L_i) + k_s (N \cdot H_i)^{n_s} \right]$$

Phong Illumination: The General Equation

Diffuse: non-shiny illumination and shadows

Specular: shiny reflections

Ambient: background illumination



Colour Considerations

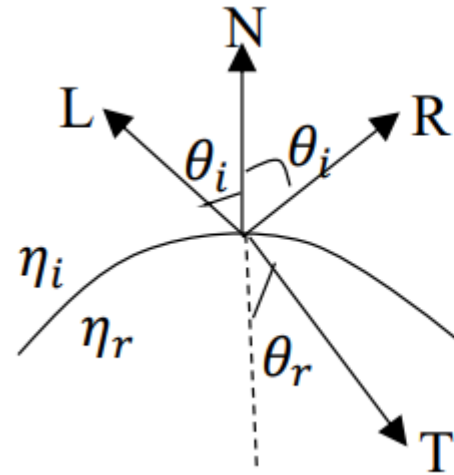
- ❑ Most graphics displays of realistic scenes are in colour. But the illumination model discussed so far considers only monochromatic lighting effects.
- ❑ To incorporate colour, we need to write the intensity equation as a function of the colour properties of the light sources and object surfaces.
- ❑ One way to set surface colors is by specifying the reflectivity coefficients as three-element vectors. The diffuse reflection coefficient vector, for example, would then have RGB components (k_dR , k_dG , k_dB)

Transparency

- ❑ A transparent surface in general produces both reflected and transmitted light
- ❑ The relative contribution of the transmitted light depends on the degree of transparency of the surface and whether any light sources are behind the transparent surface
- ❑ Both diffuse and specular transmission can take place at the surfaces of a transparent object

According to Snell's Law

$$\sin \theta_r = \frac{\eta_i}{\eta_r} \sin \theta_i$$



θ_i =angle of incidence

η_i =index of incidence

θ_r =angle of refraction

η_r =index of refraction

Fig: Reflection direction R and refraction direction T for a ray of light incident upon a surface with index of refraction η_r

Total surface intensity is calculated as

$$I = (1 - k_t)I_{reflect} + k_t I_{trans}$$

Where,

$I_{reflect}$ = reflected intensity

I_{trans} = transmitted intensity

k_t = transparency coefficient

$(1 - k_t)$ = opacity factor

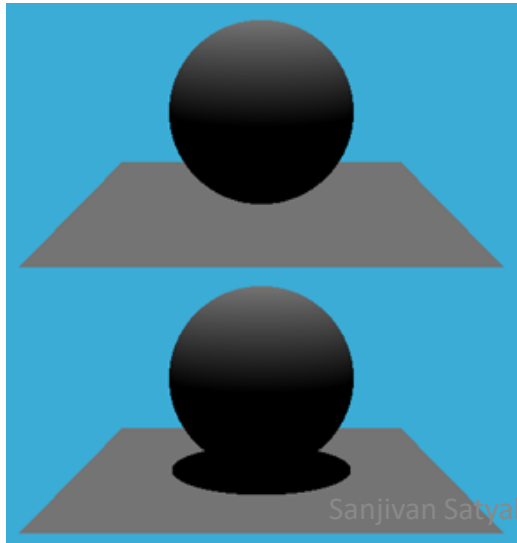
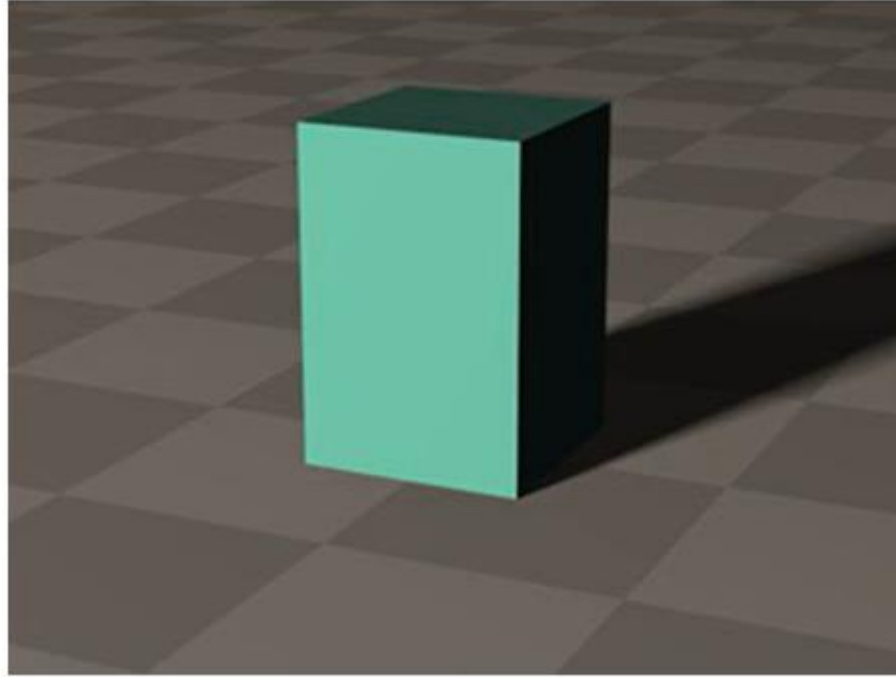
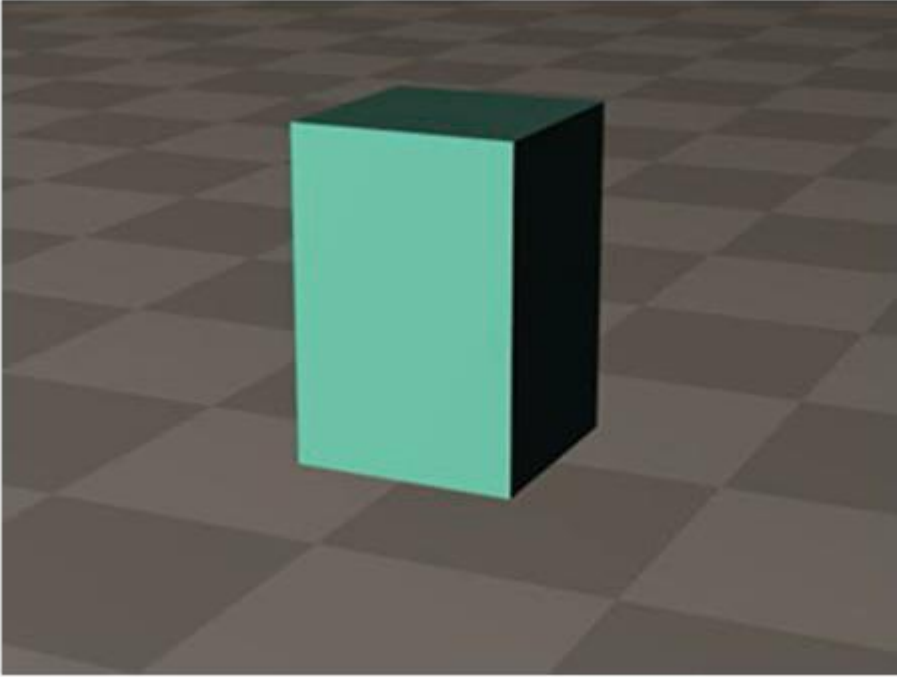
For highly transparent objects k_t is near 1.

For opaque objects k_t is near 0.

Shadows

- ❑ Hidden surface method can be used to locate area where light sources produce shadows.
- ❑ By applying a hidden surface method with a light source at the view position, we can determine which surface sections cannot be seen from the light source.
- ❑ Once we have determined the shadow area for all light sources, the shadows could be treated as surface patterns and store in pattern arrays.
- ❑ Shadow patterns generated by a hidden surface method are valid for any selected viewing position, as long as the light source positions are not changed.
- ❑ Shadow can help to create realism. Without it, a cup, e.g., on a table may look as if the cup is floating in the air above the table.

Shadows tell us about the relative locations and motions of objects



Polygon Rendering Method

- ❑ The surface rendering method is used to find the total light intensity on the whole surface
- ❑ The standard objects that are formed by the polygon surfaces are rendered by the application of polygon rendering method
- ❑ **Two ways** of polygon surface rendering:
 1. Single intensity for all points in a polygon
 2. Interpolation of intensities for each point in a polygon

Methods:

- a) **Constant intensity shading method**
- b) **Gouraud shading method (Intensity Interpolation)**
- c) **Phong shading method (Normal vector interpolation)**
- d) **Fast Phong Shading**

Polygon Rendering Methods

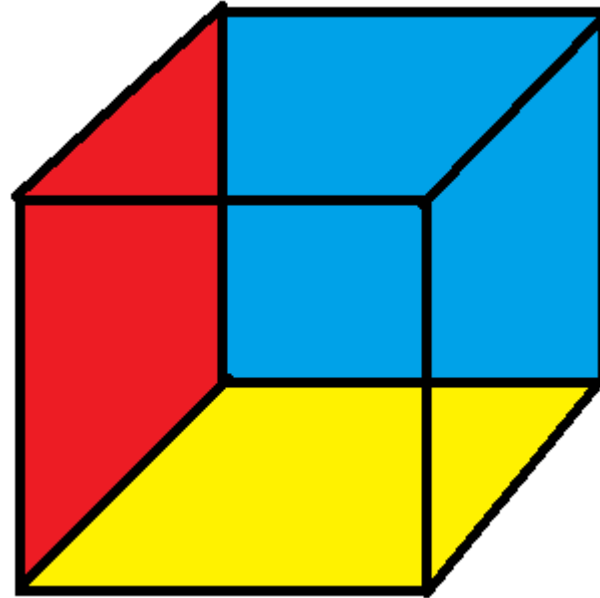
- **Constant intensity rendering:**
 - No interpolation.
 - Also called "flat shading"
- **Intensity interpolation rendering:**
 - "Gouraud shading"
 - Based on intensity interpolation of vertices
- **Normal vector interpolation rendering:**
 - "Phong shading"
 - Based on interpolation of normal vectors of vertices



1. Constant Intensity Shading (Flat Shading)Method

- ❑ A fast and simple method for rendering an object with polygon surfaces
- ❑ All the points over the surface of the polygon are displayed with same intensity value
- ❑ In this method, a single intensity is calculated for each polygon
 - ❑ Determine color for center of polygon
- ❑ And the intensity is applied to all the points of surface of polygon.
 - ❑ Fill the polygon with a constant color.

Constant Intensity Shading (Flat Shading)Method

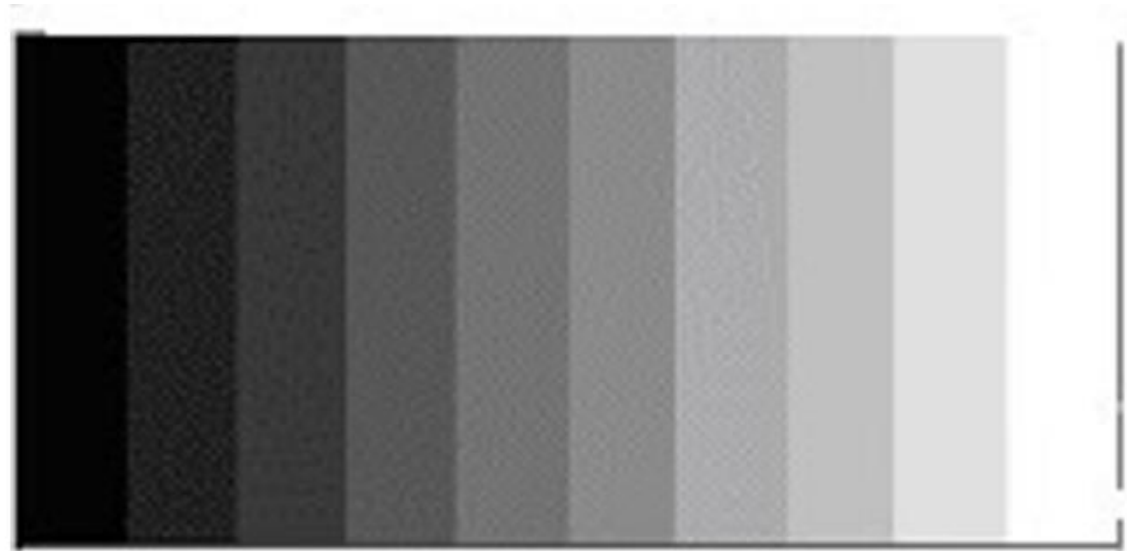
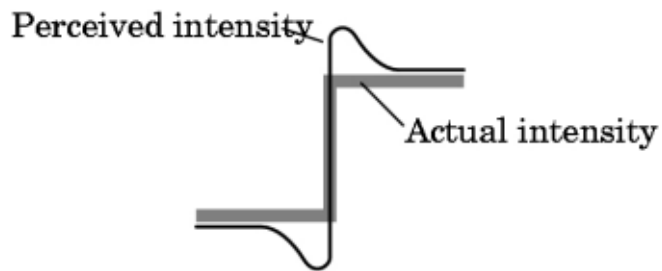


- ❑ Constant shading is useful for quickly displaying the general appearance of a curved surfaces. This approach is valid if:
 - Object consists of planar faces, and
 - Light sources are far away, and
 - Eye point is far away, or
 - Polygons are about a pixel in size.
- ❑ Efficient, but shows too much (too abrupt) shading difference between adjacent polygons.
- ❑ Highlights not visible, Facetted appearance, increased by Mach banding effect

Mach band Effect

- The **Mach band effect** is an optical illusion that occurs due to the way our visual system interprets transitions between different brightness levels. In computer graphics, it refers to the perception of exaggerated differences in intensity at the boundaries of regions where brightness changes.
- This effect happens when there is a sharp transition between light and dark areas, and the human visual system perceives more pronounced "bands" of brightness than actually exist. These "bands" aren't real; they're a result of the way the brain processes visual information, amplifying small differences in light intensity at the edges of objects or regions.

□ **Mach bands** can be reduced by dividing the surface into a greater number of polygon faces or Phong shading (*requires more calculation*)



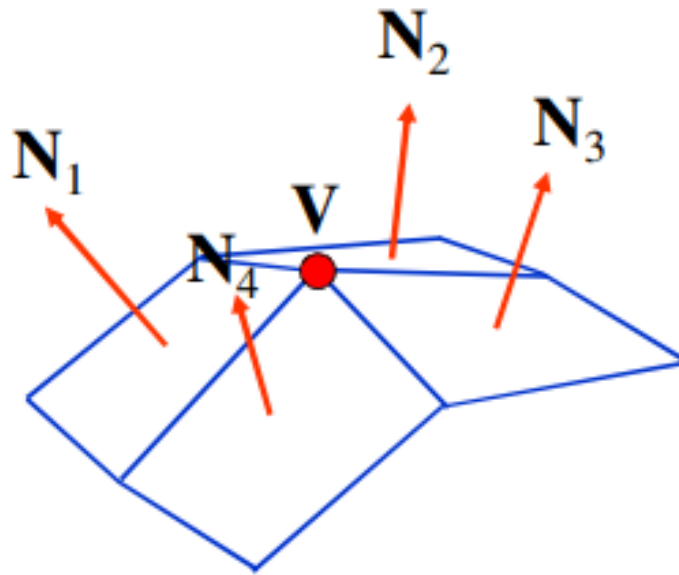
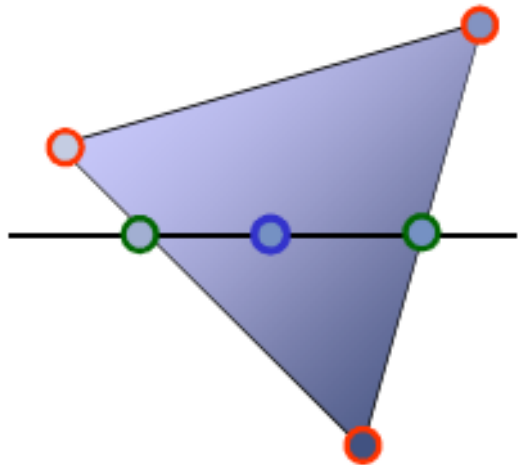
2. Gouraud Shading

- ❑ Developed by Henri Gouraud.
- ❑ It renders the polygon surface by linearly interpolating intensity values across the surface
- ❑ Colour is computed once per vertex using the local illumination model
- ❑ Intensity values for each polygon are matched with the values of adjacent polygon along the common edge, thus eliminating the intensity discontinuities that occur in flat shading



PROCEDURE

1. Determine the average unit normal vector at each polygon vertex
2. Calculate each of the vertex intensities by applying an illumination model
3. Linearly interpolate the vertex intensities over the polygon surface



$$\mathbf{N}_V = \frac{\sum_{k=1}^n \mathbf{N}_k}{\left| \sum_{k=1}^n \mathbf{N}_k \right|}$$

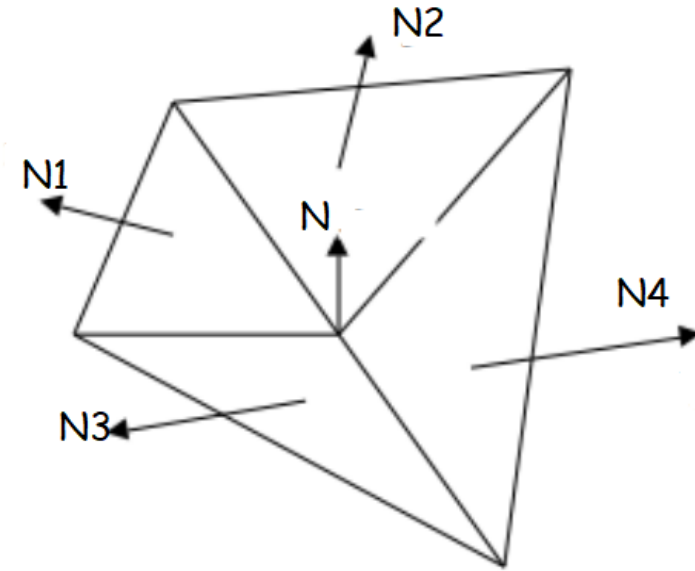
Gouraud Shading...

- The intensity value is calculated once for each vertex of a polygon.
- Each polygon has one normal vector per vertex, the color of each vertex is computed and then interpolated across the surface of the polygon.
- The interpolation of color values can cause bright or dark intensity streaks, called the Mach-bands, to appear on the surface
- Calculation for each polygon surfaces
 - Determine the average unit normal vector at each vertex. At each polygon vertex, we obtain a normal vertex by averaging the surface normals of all polygons sharing the vertex as:

$$\mathbf{N}_v = \frac{\sum_{k=1}^n \mathbf{N}_k}{\left| \sum_{k=1}^n \mathbf{N}_k \right|}$$

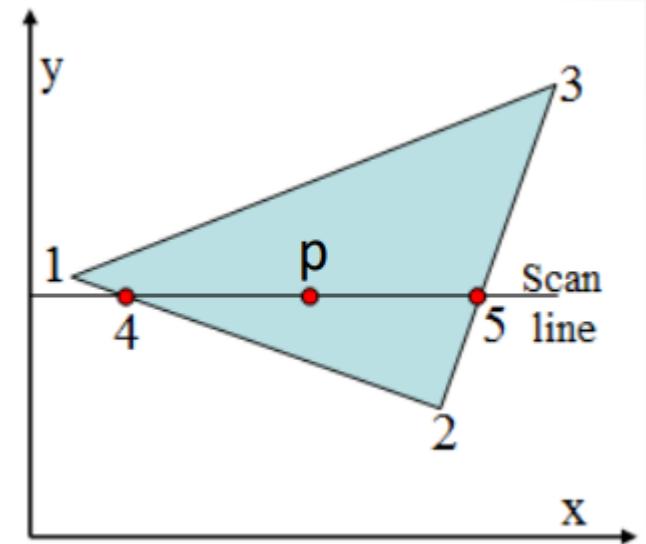
$$N_v = \frac{N_1 + N_2 + N_3 + N_4}{|N_1 + N_2 + N_3 + N_4|}$$

Where N_v is normal vector at a vertex sharing Four surfaces as in figure.



- For Gouraud shading, the intensity at point 4 is linearly interpolated from the intensities at vertices 1 and 2.
- The intensity at point 5 is linearly interpolated from intensities at vertices 2 and 3.
- An interior point p is then assigned an intensity value that is linearly interpolated from intensities at positions 4 and 5.

$$I_4 = \frac{y_4 - y_2}{y_1 - y_2} I_1 + \frac{y_1 - y_4}{y_1 - y_2} I_2 \quad I_p = \frac{x_5 - x_p}{x_5 - x_4} I_4 + \frac{x_p - x_4}{x_5 - x_4} I_5$$



Advantages:

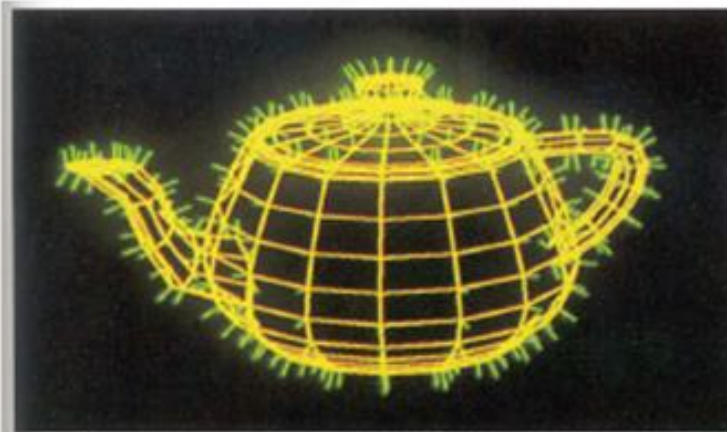
- ❑ Removes discontinuities of intensity at the edge compared to constant shading model

Gouraud Shading Problems

- ❑ still not smooth
- ❑ Lighting in the polygon interior is inaccurate
- ❑ Mach band effect
- ❑ interpolates linearly and so can make the highlight much bigger
- ❑ can miss highlights that occur in the middle of a polygon

Solution: Could be reduced by dividing the surface into large number of polygons or by using other methods, such as **Phong shading**

Wireframe



Flat



Gouraud



Phong

3. Phong Shading

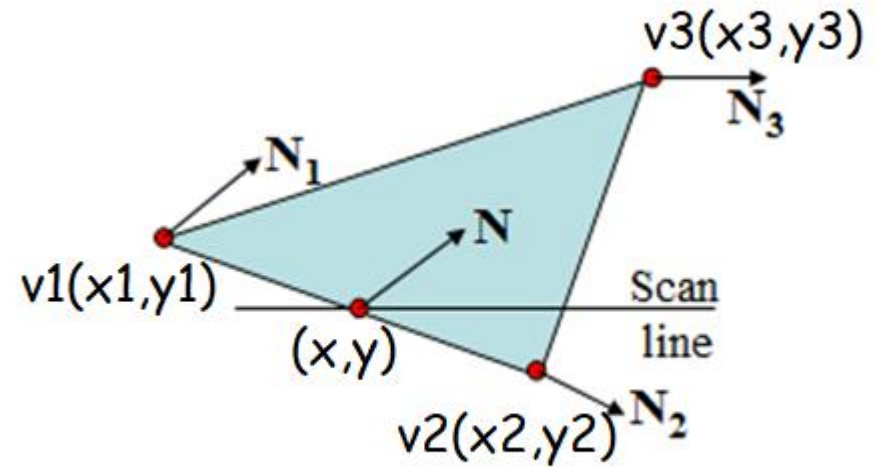
- ❑ Developed by Phong Bui Tuong.
- ❑ A more accurate method for rendering a polygon surface is Phong shading, or normal vector interpolation shading
- ❑ It renders the polygon surface by linearly interpolating **normal vector** across the each surface point
- ❑ It displays more realistic highlights on a surface and greatly reduces the Mach-band effect

A polygon surface is rendered using Phong shading by carrying out the **following steps**:

1. Determine the average unit normal vector at each polygon vertex.
2. Linearly interpolate the **vertex normals** over the surface of the polygon
3. Apply an illumination model along each scan line to calculate projected pixel intensities for the surface points.

The normal vector $\mathbf{N}$ for the scan-line intersection point along the edge between vertices 1 and 2 can be obtained by vertically interpolating between edge endpoint normal:

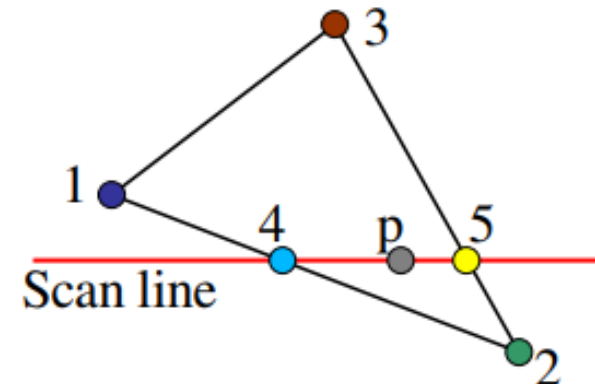
$$\mathbf{N} = \frac{y - y_2}{y_1 - y_2} \mathbf{N}_1 + \frac{y_1 - y}{y_1 - y_2} \mathbf{N}_2$$



- ❑ Expensive since it requires calculation of intensity for each pixel.
- ❑ Accelerations based on Taylor expansion exists but still very expensive compared to Gouraud

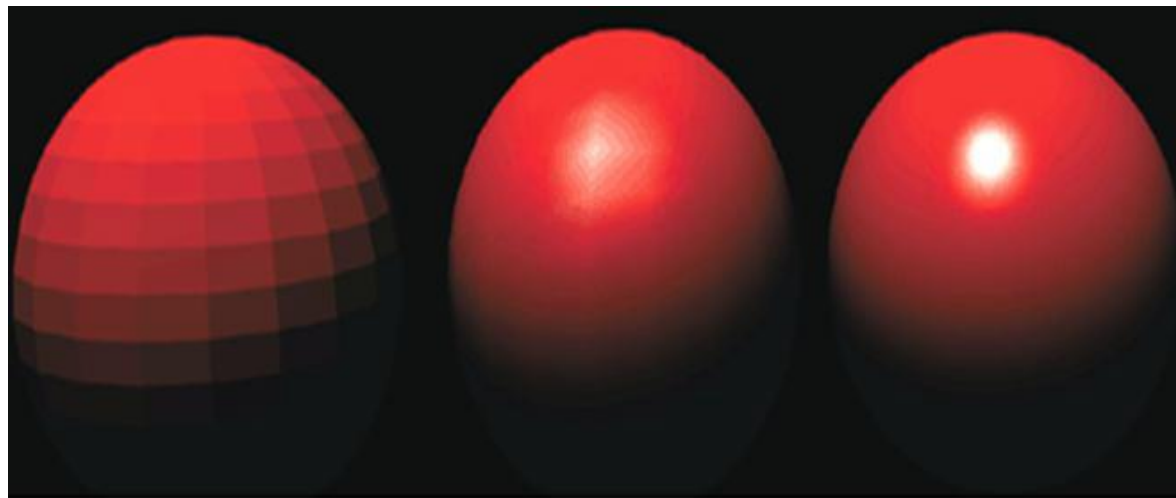
$$N_4 = \frac{y_4 - y_2}{y_1 - y_2} N_1 + \frac{y_1 - y_4}{y_1 - y_2} N_2$$

$$N_P = \frac{x_5 - x_P}{x_5 - x_4} N_4 + \frac{x_P - x_4}{x_5 - x_4} N_5$$



Phong Shading...

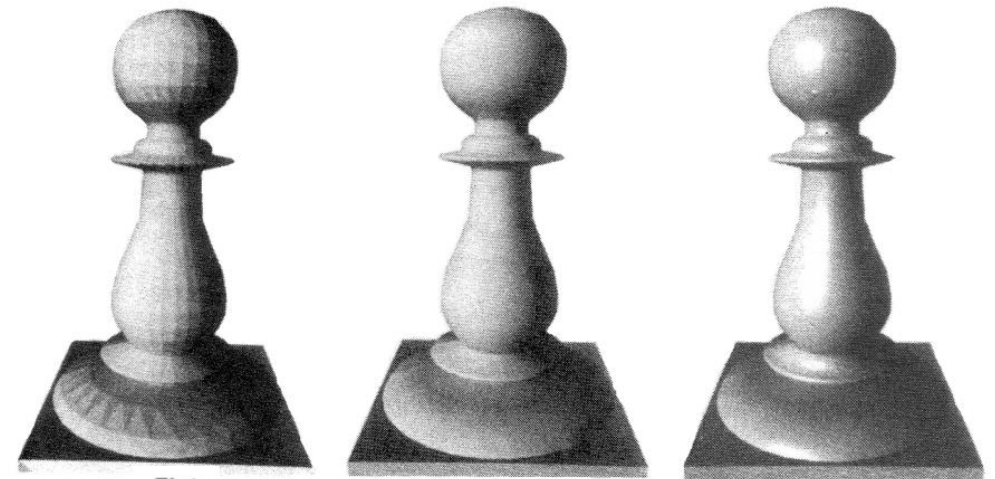
- Incremental methods are used to evaluate normal between scan lines and along each individual scan line (as in Gouraud)
- At each pixel position along a scan line the illumination model is applied to determine the surface intensity at that point
- Intensity calculations using an **approximated normal vector** at each point along the scan line produce more accurate results than the **direct interpolation of intensities**, as in Gouraud shading but it requires considerable more calculations.



Flat

Gouraud

Phong



Flat

Gouraud

Phong

Gouraud versus Phong Shading

Key differences:

1. Calculation point:

- Gouraud calculates lighting at polygon vertices, Phong calculates lighting at each pixel.

2. Interpolation:

- Gouraud interpolates color values across the polygon, Phong interpolates surface normals across the polygon.

3. Visual quality:

- Gouraud can appear faceted, especially on curved surfaces, while Phong produces smoother shading with more accurate specular highlights.

4. Computational cost:

- Gouraud is computationally cheaper due to fewer calculations per pixel, while Phong is more computationally expensive.

When to use which:

5. Gouraud shading:

- Suitable for applications where fast rendering is prioritized, like low-poly models or situations where a slightly less realistic appearance is acceptable.

6. Phong shading:

- Preferred for high-quality rendering where accurate specular highlights and smooth shading are important, especially for complex or reflective surfaces

Flat shading	Gouraud shading	Phong shading
Computes illumination once per polygon and apply it to whole polygon.	Computes illumination at vertices and interpolates.	Applies illumination at every point of polygon surface.
Creates discontinuous in color.	Interpolates colors along edges and scan line.	Interpolates normal instead of colors.
Problems of Mach bands.	Handles Mach bands problem found in flat shading.	Removes Mach bands completely.
Low cost.	Not so expensive.	More expensive than gouraud shading.
Requires very less processing and is fast in time.	Required moderate processing time.	Requires complex processing and is slower but is more efficient as compared to other shading methods.
Lighting equation used once per polygon.	Lighting equation used at each vertex.	Lighting equation used at each pixel.

4. Fast Phong Shading

- **Fast Phong** shading is a technique in computer graphics that speeds up the Phong shading algorithm by using **Taylor series expansions** and triangular surface patches. It's used to reduce the time it takes to render images.

How it works

- Fast Phong shading uses Taylor series expansions to approximate the calculations for illumination and normal vectors.
- It uses triangular surface patches to express the surface normal at any point on a triangle.
- It uses forward differences to evaluate the diffuse reflection calculation.

Surface normal at any point (x, y) over a triangle as; $N = Ax + By + C$

A, B, C are determined from the three vertex equations:

$$N_k = Ax_k + By_k + C, \quad k = 1, 2, 3$$

With (x_k, y_k) denoting a vertex position.

Omitting the reflectivity and attenuation parameters,

$$I_{diff}(x, y) = \frac{L \cdot N}{|L||N|} = \frac{L \cdot (Ax + By + C)}{|L||Ax + By + C|} = \frac{(L \cdot A)x + (L \cdot B)y + L \cdot C}{|L||Ax + By + C|}$$

Where , L light source

We can write this,

$$I_{diff}(x, y) = \frac{ax + by + c}{(dx^2 + exy + fy^2 + gx + hy + i)^{1/2}} \dots \dots \dots (i)$$

Where, a, b, c, d are used to represent the various dot product. For e.g.

$$a = \frac{L \cdot A}{|L|}$$

Finally, denominator in eq. (i) can be expressed as a Taylor-series expansion and retain terms up to second degree in x & y. This yields

$$I_{diff}(x, y) = T_5x^2 + T_4xy + T_3y^2 + T_2x + T_1y + T_0$$

Where, each T_k is a function of parameters a, b, c, and so forth.

Advantages

- ❑ Fast Phong shading is faster than normal Phong shading.
- ❑ It can be extended to include specular reflections and support polygons other than triangles.

Thank you